

PROGRESS UPDATE



AI-Based Energy Reconstruction for the ATLAS Tile Calorimeter at HL-LHC

Progress about the a calibrated teacher, a full distillation study, and the path to the FPGA

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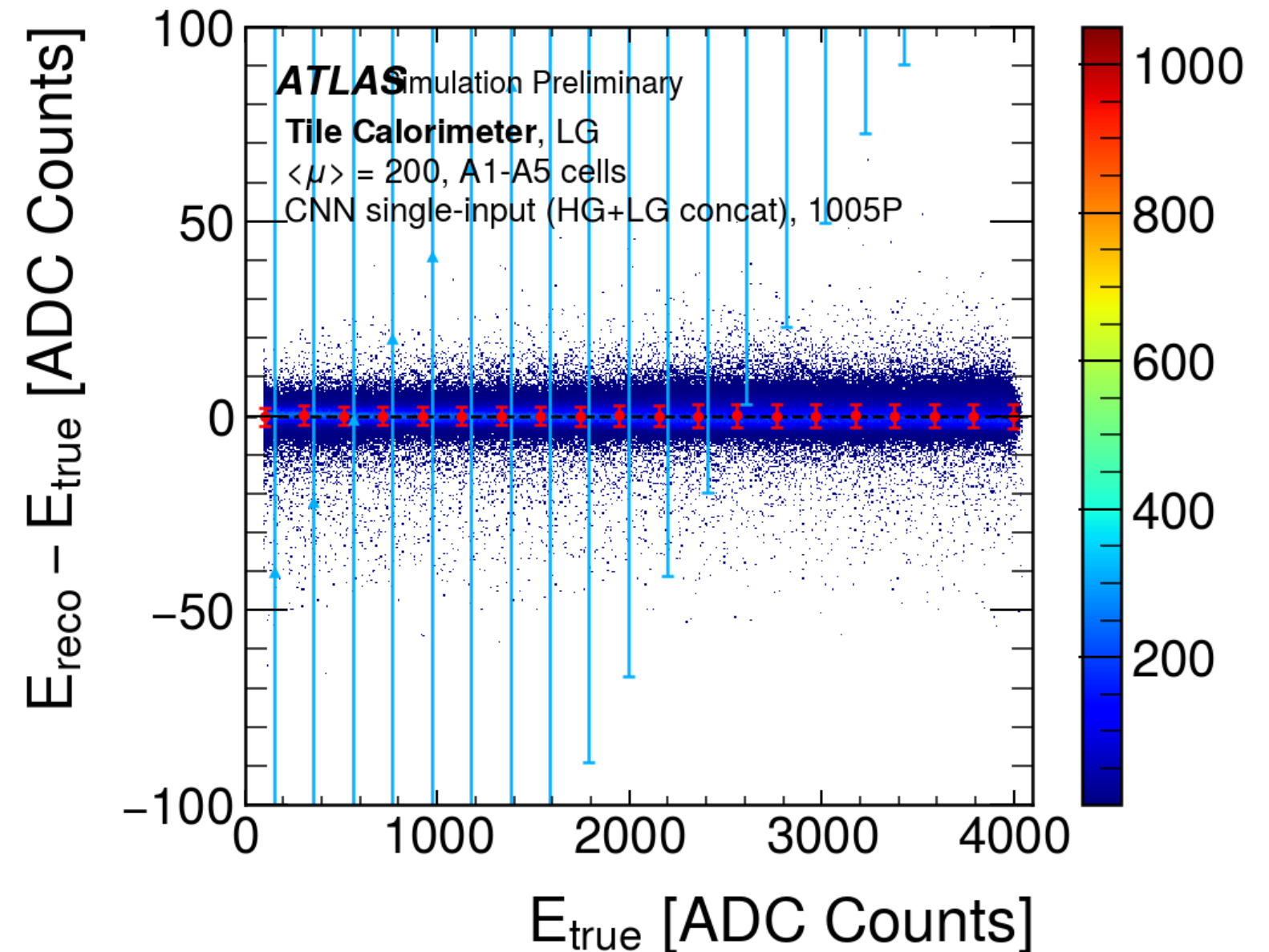
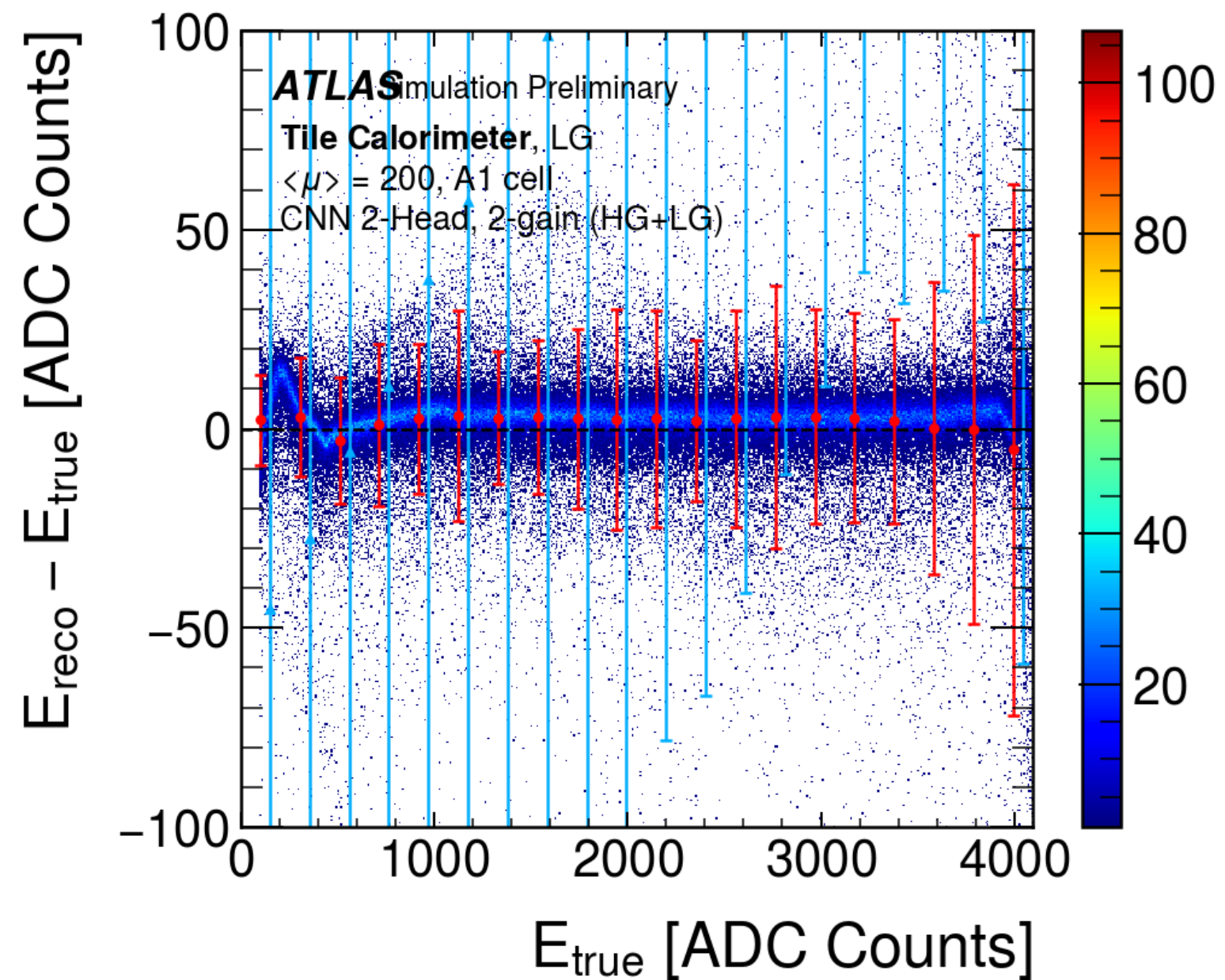
Recap: What We Agreed Last Time

- Adopt the single concatenated dual-gain input $\mathbf{S} = \mathbf{HG} + 4096*\mathbf{LG}$ with one target, shared exactly by teacher and student.
- Build a genuinely good teacher first, growing it in ~10% steps.
- Report resolution as the **per-bin mean +/- standard deviation vs energy** ("the band"), not a single global number.
- Train multiple cells together (A1-A5), compare 8 vs 9 bunch crossings.

The Unified Concatenated Input

- Switched to single input **$S = HG + 4096 * LG$** (HG low 12 bits, LG high 12 bits), one target ene_lo, HG view = x40.
- FPGA-friendly activation:- **LeakyReLU(0.25)** = 2^{-2} bit-shift (parameter-free, replaces PReLU).
- Fixed the low-gain "S-shape", LG mean now **flat at 0 with -3 ADC width** across the whole range.
- Engineering enabler:- data kept GPU-resident, epoch time **-5.5 min -> -20 s**, bit-for-bit identical.

The Unified Concatenated Input

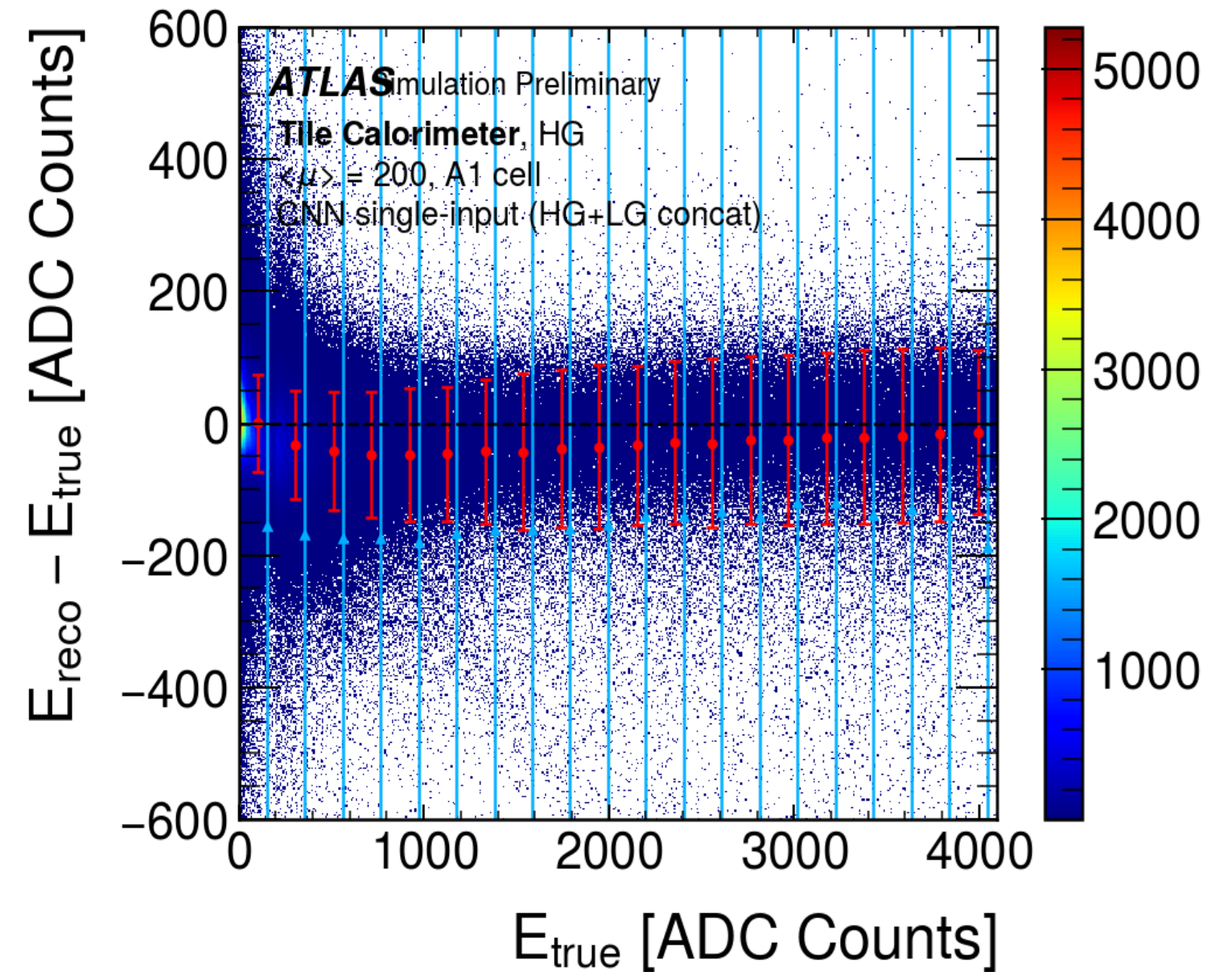
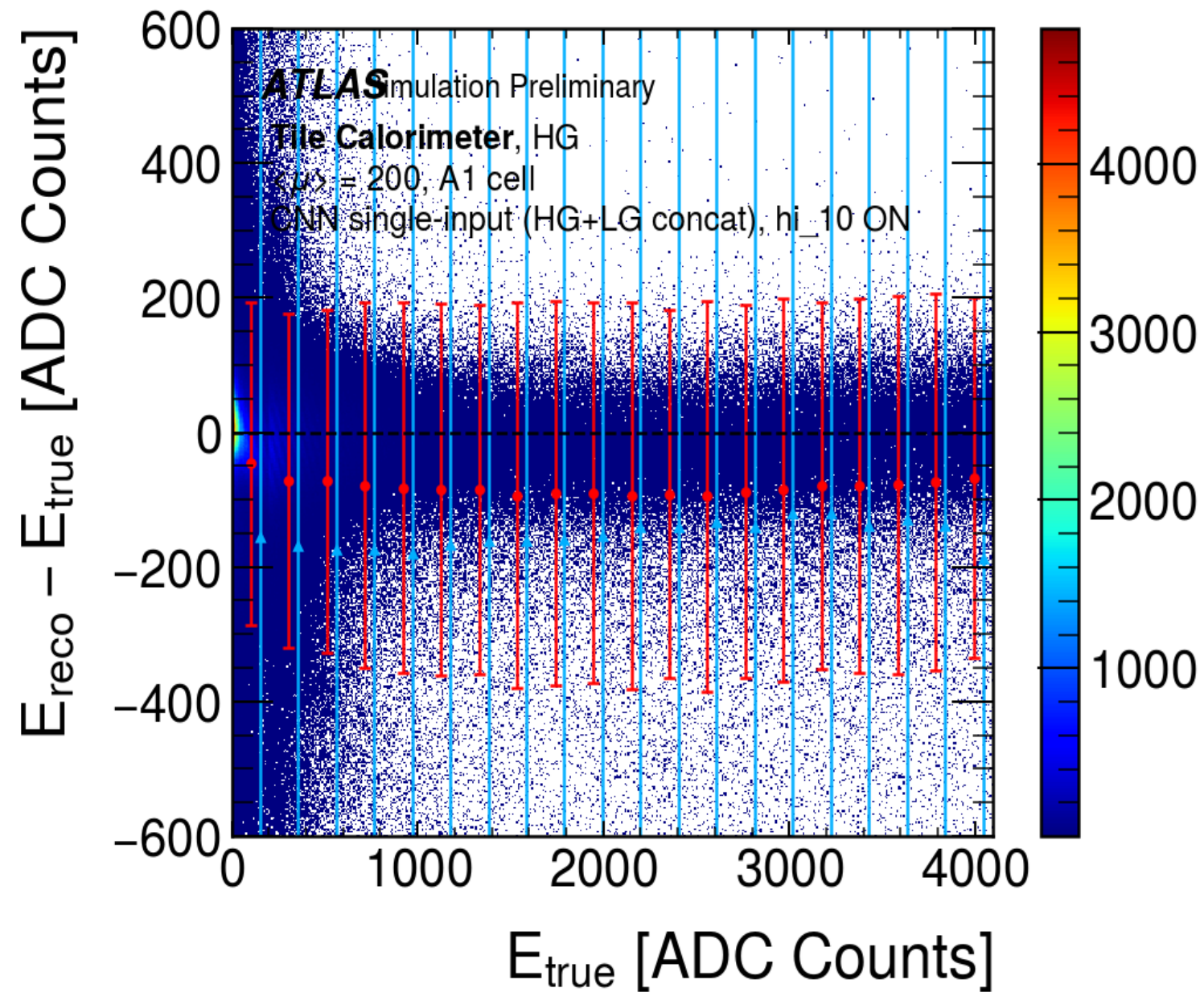


LG band before vs after the unified input
(S-shape removed)

Teacher Design Tests (incl. dead ends)

- "Drop if any HG sample < 10 ADC" cut-: WITH it the band **~triples** and mean drifts negative -> **kept OFF**.
- **8 vs 9 BC**:- LG identical, HG ~5% wider -> stay on 9 BC, keep 25 ns in reserve.
- **Multi-cell A1-A5**:- resolution matches single-cell A1 -> one model covers all A-layer cells.
- **Dead end**:- energy-weighted loss put weight on high-energy region, made HG slightly worse -> dropped.

Teacher Design Tests (incl. dead ends)

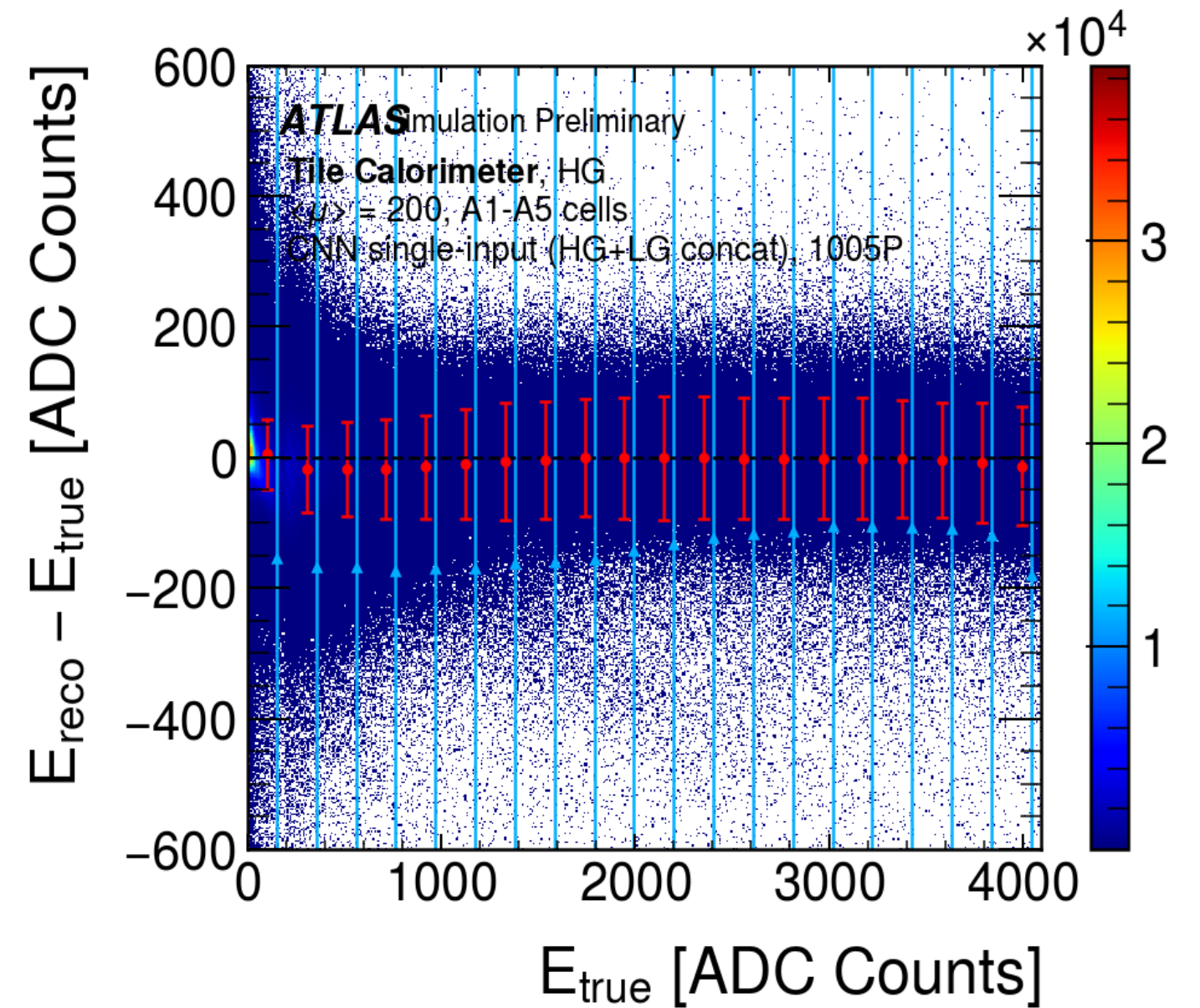
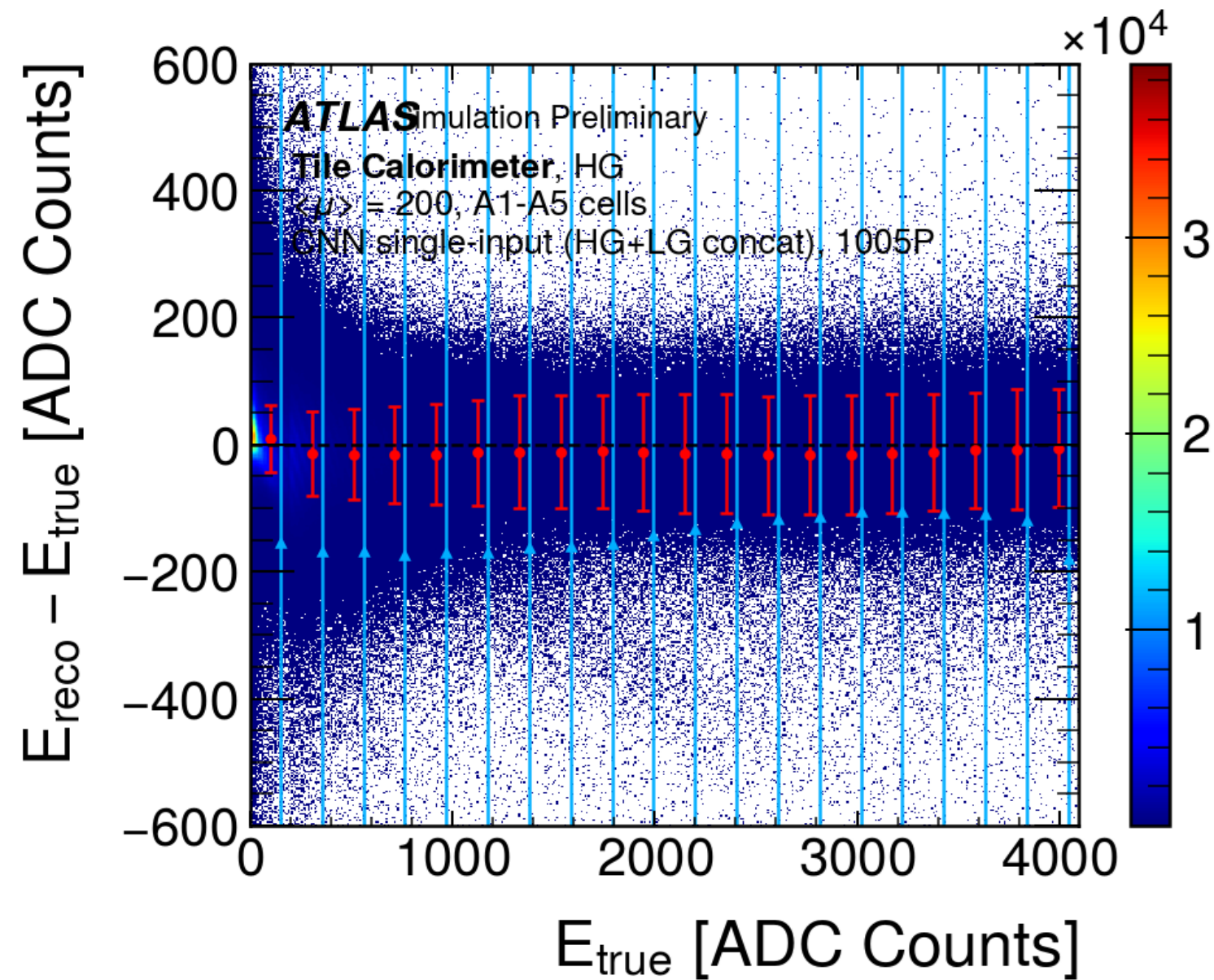


with-cut vs without-cut HG band
on the common test

The HG Offset and Post-hoc Calibration

- Raw **--20 ADC HG offset** from single LG target:– tiny --0.5 ADC bias amplified x40 on HG view. (Reference uses separate per-gain models.)
- **Calibration:**– measure bias vs prediction on validation, subtract on LG target, one curve fixes both gains. Offset removed, **sigma UNCHANGED**.
- Only residual:– very-low-energy HG dip (fundamental, also in reference and ATLAS LAr CNN plots).
- **Honesty check:**– uniform-bin calibration only shifts mean (sigma identical), finer quantile version artificially shrank sigma -> **rejected**.

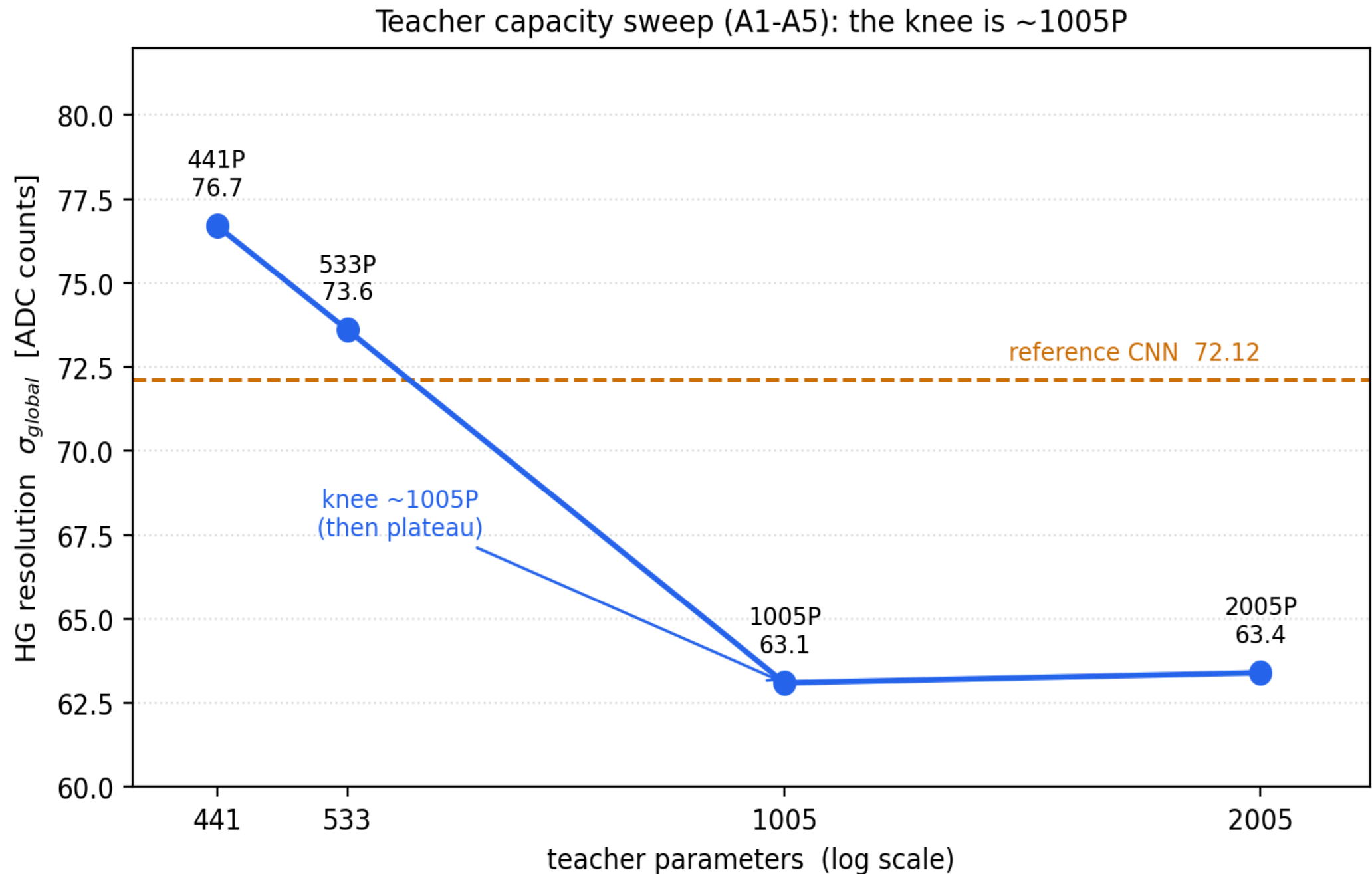
The HG Offset and Post-hoc Calibration



Raw vs calibrated HG band
(offset removed, sigma unchanged)

Parameter Sweep -> the 1005P Teacher

- Earlier 441->533 hitting a wall looked like the floor, with A1-A5 data + speedup, larger models became testable.
- HG global sigma vs size:- **-441P -77 -> -533P -74 -> -1005P -63 -> -2005P -63.**
- At -1005P HG tightened **-15% in every bin**, then plateaued at -2000 -> knee is -1000 parameters, LG stayed -2.7 ADC.
- Confirms:- extra parameters only help once extra input data (cells) is added. **The 1005P CNN is the teacher.**



HG global sigma vs parameter count
(441 / 533 / 1005 / 2005)

Teacher Results (1005P, A1-A5, calibrated)

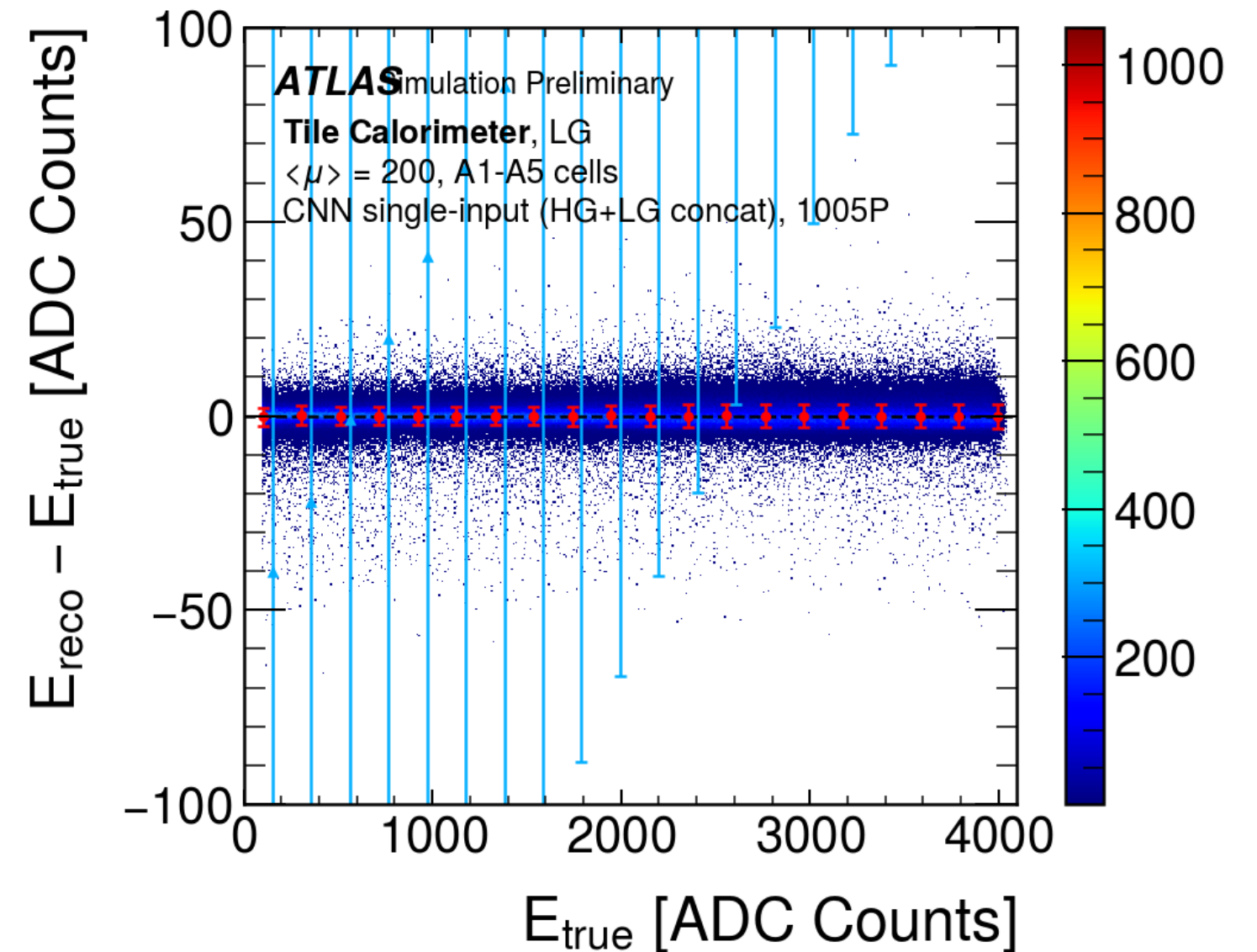
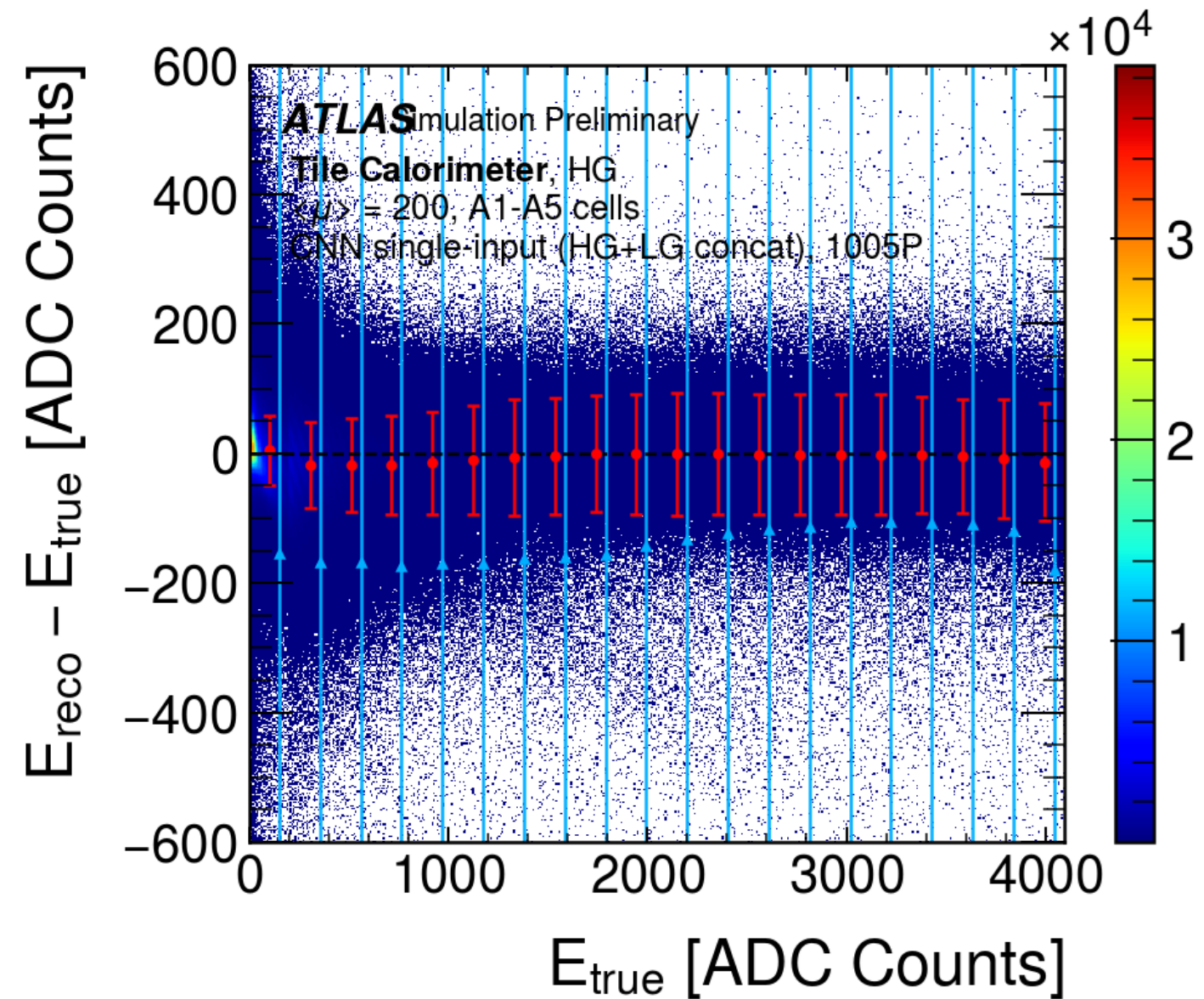
Conv1d(1->18,k5) -> Conv1d(18->14,k3) -> Linear(126->1), LeakyReLU(0.25); 1005 params | ificuw/tilecal!16 (merged)

- HG band:- $\sim \pm 53$ ADC at low energy widening to $\sim \pm 93$ toward HG/LG boundary, mean flat, slope ~ 0.99 , $R^2 \sim 0.99$.
- High-energy HG end does not move with more params or data -> **irreducible pile-up floor**, not capacity limit.

Resolution summary (sigma, ADC counts)

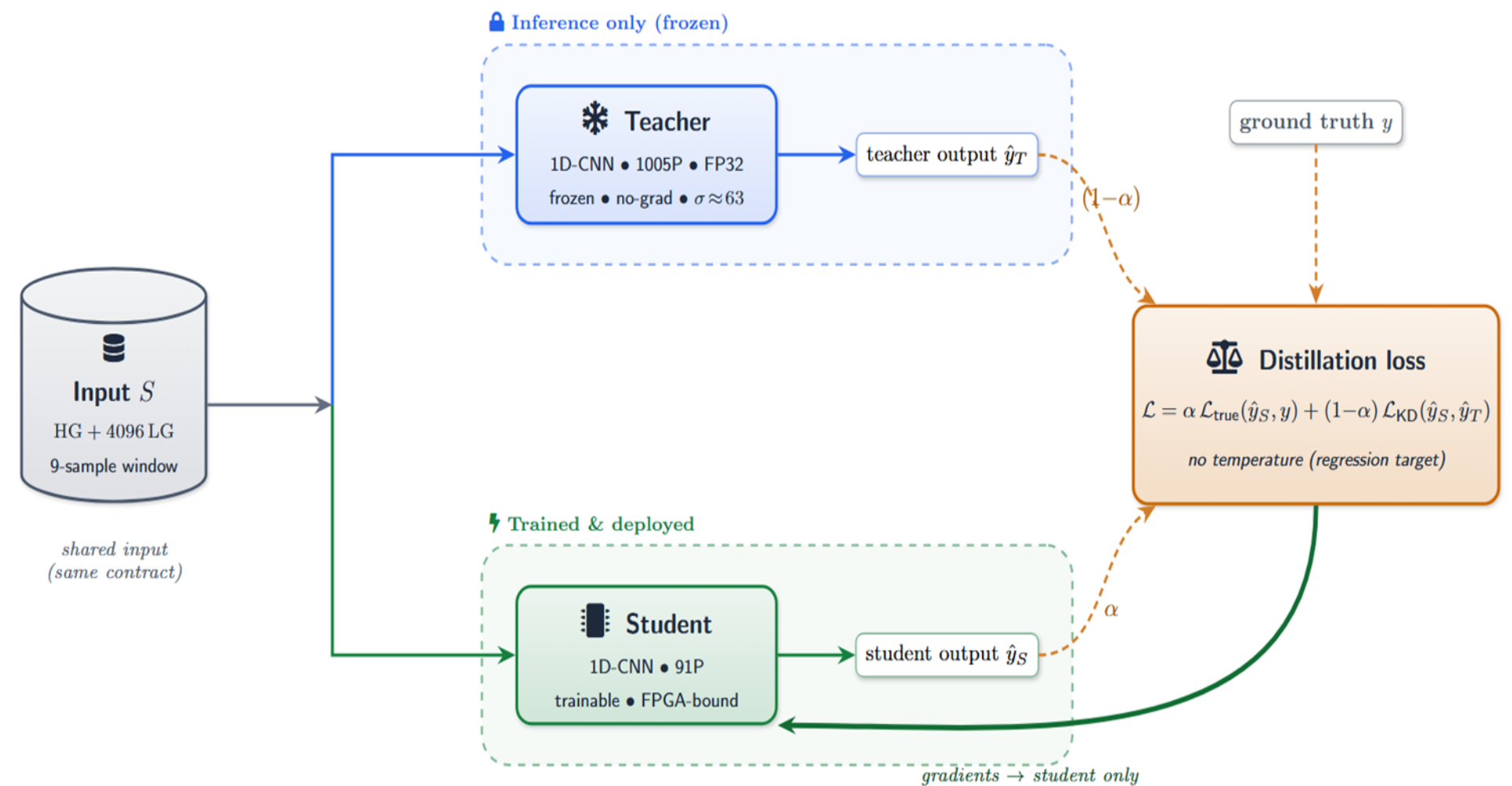
Model	HG sigma (global)	LG sigma
Teacher (1005P, ours)	~ 63	~ 2.7
IFIC ref. CNN (147P, HG-only)	72.12	8.36
Optimal Filtering (context)	$\sim 900-1000$	~ 250

Teacher Results (1005P, A1-A5, calibrated)



Knowledge Distillation, Framing & Setup

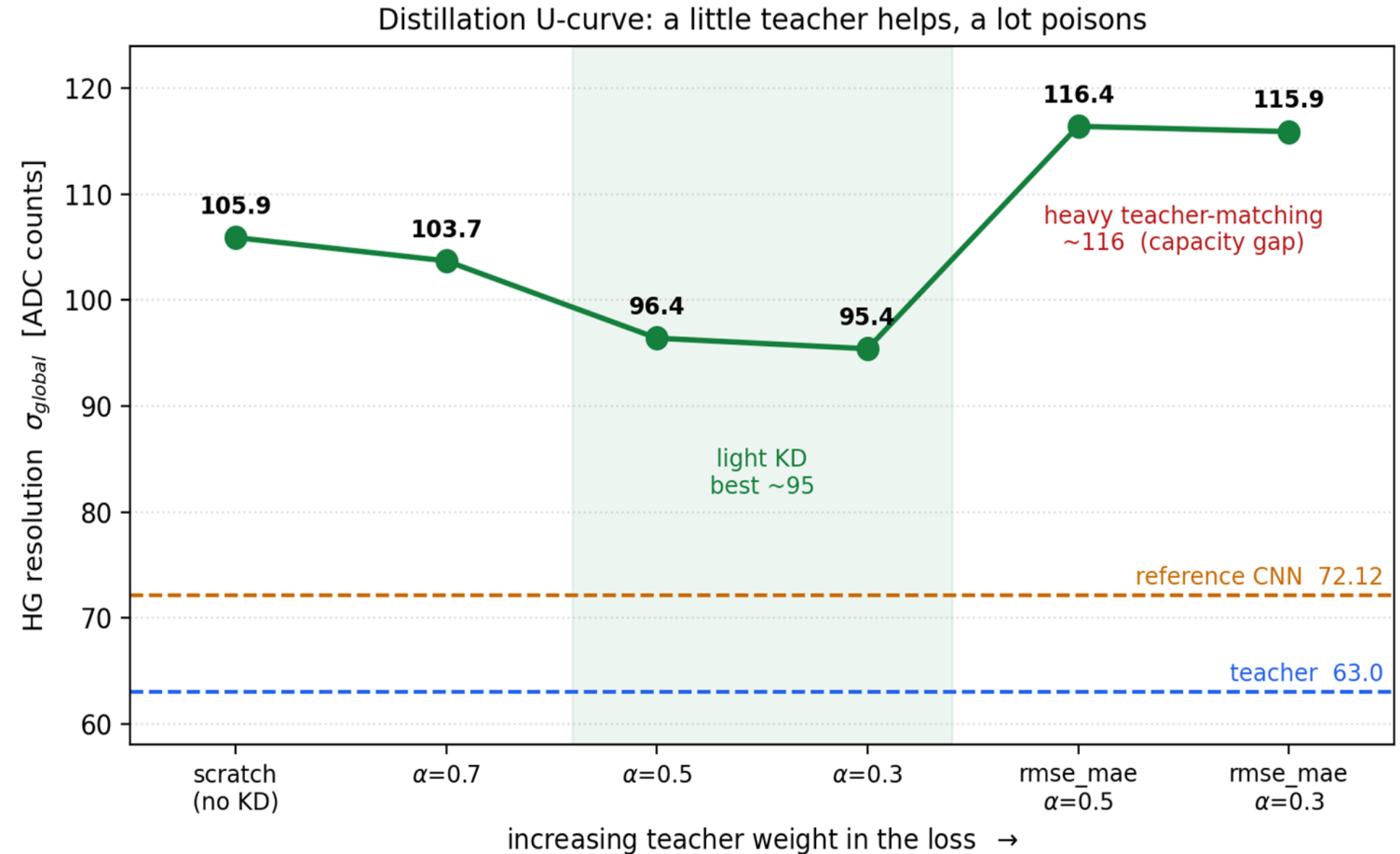
- Teacher sits at pile-up noise floor -> distillation is **variance management**, not accuracy transfer, copying too literally teaches noise reproduction,
- Loss:- $L = \alpha * L_{\text{true}} + (1-\alpha) * L_{\text{KD}}$. No temperature (classification construct; this is regression).
- Student shares teacher's exact I/O and calibration; students trained were **91- and 99-parameter 1D-CNNs**.
- KD hook is additive:- load frozen teacher, one extra forward pass per batch, add teacher term. Early-stopping stays on true-label loss.



Teacher -> student distillation
(frozen teacher, weighted loss)

Distillation Exp 1, Teacher Weight (alpha sweep)

- Swept effective teacher weight across full range, read per-bin band of each student (91P CNN).
- Clear **U-curve** in HG sigma:- scratch -106 -> light KD -95 -> heavy matching -116 (worse than scratch).
- Heavy teacher-matching makes student fit TRUTH worse than no-teacher -> **curse of capacity gap** (91P cannot represent 1005P function).
- **Takeaway**:- light distillation helps (-10% over scratch), binding limit is student capacity, not tuning.

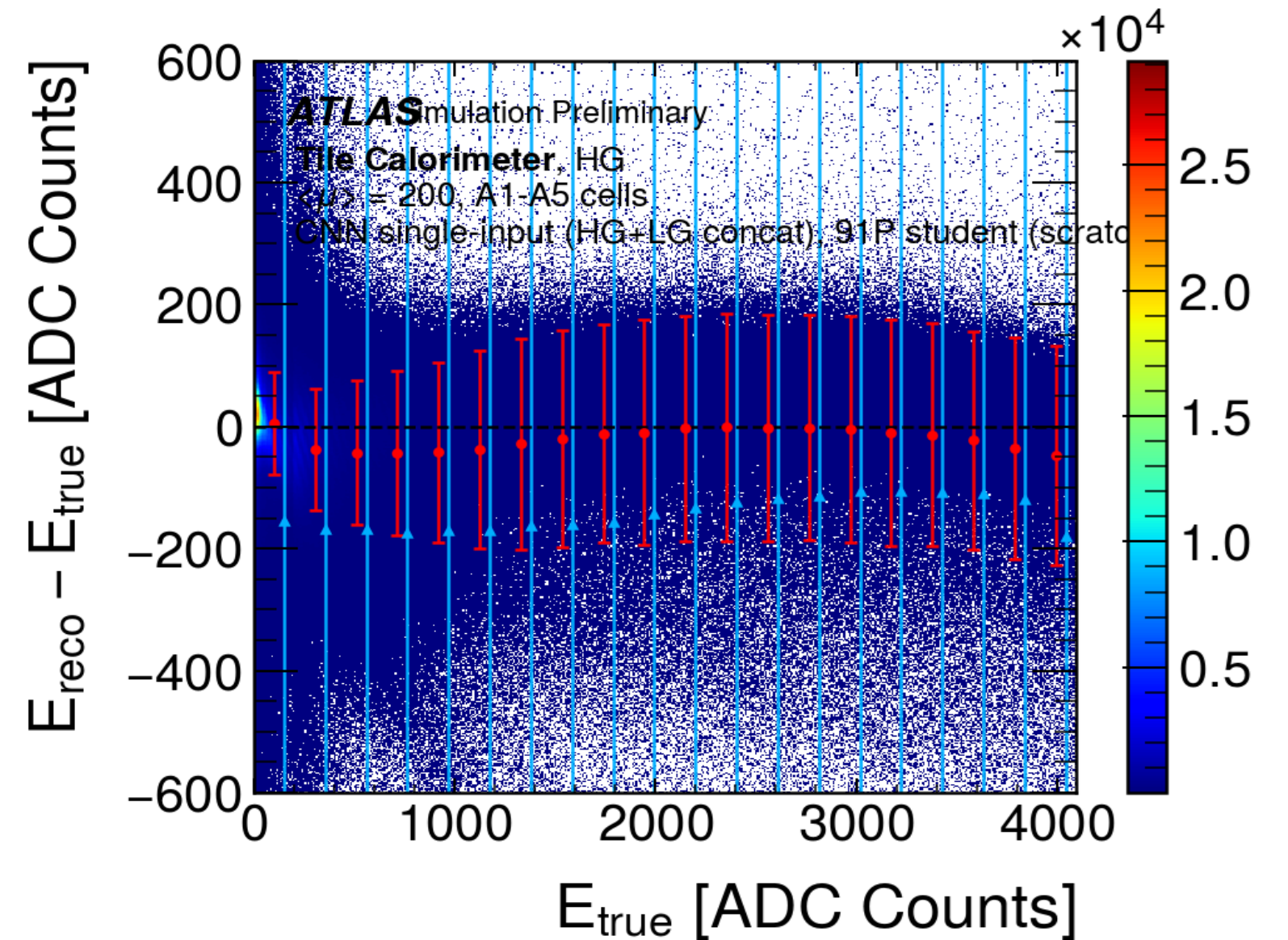
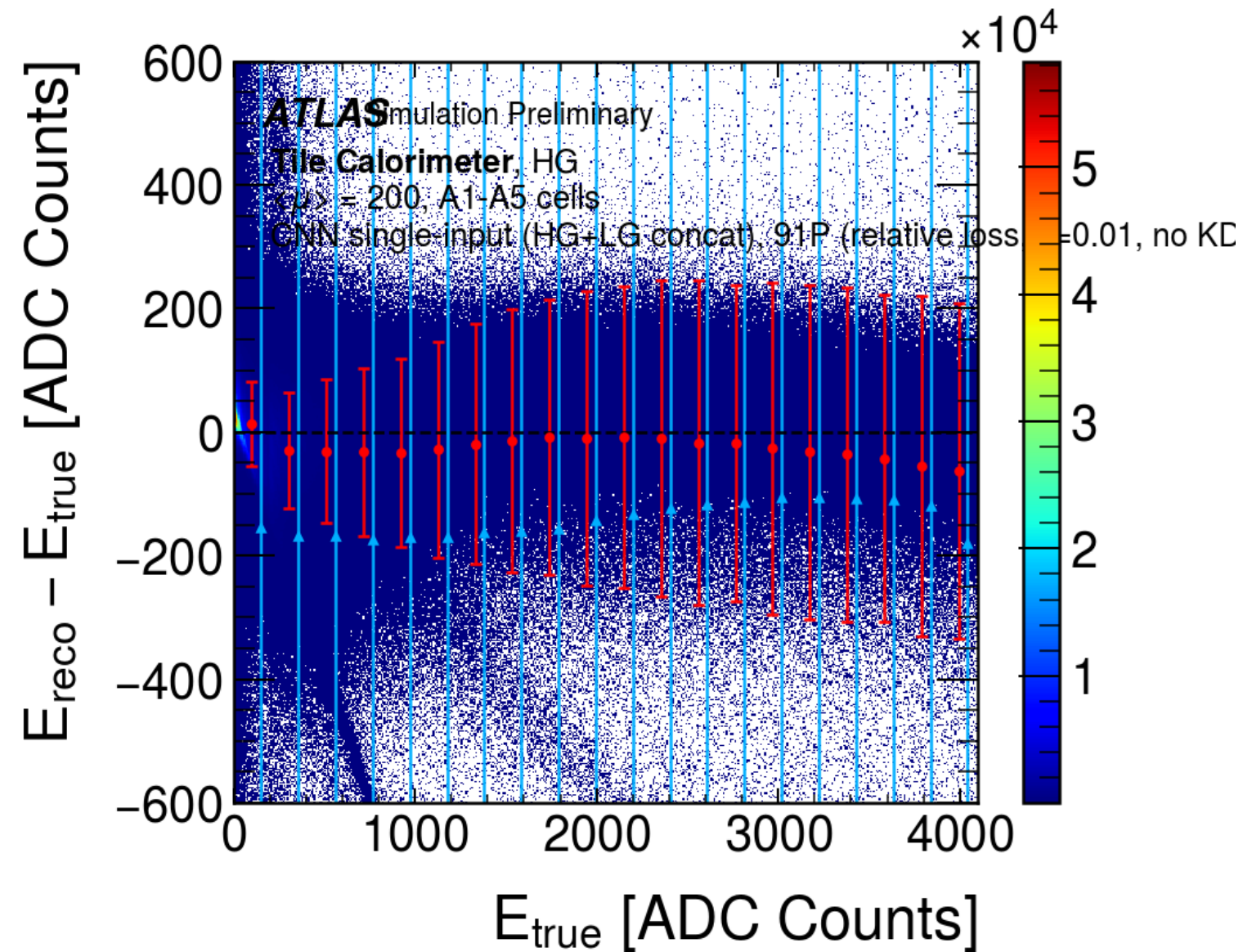


HG sigma vs effective teacher weight
(U-curve:- 106 -> 95 -> 116)

Distillation Exp 2, Loss Shaping (negative result)

- **Hypothesis:**– re-weight loss toward low energy (fractional resolution $\sigma(E)/E$) to tighten HG band where widest.
- **Result, worse, not better.** HG global ~ 110 , per-bin band blew up (bin-mean ~ 213 vs ~ 165 scratch); LG also degraded.
- **Why:**– relative/absolute-error loss optimises median, while sigma is set by mean, de-weights high energy, so those bins widened sharply.
- **Conclusion:**– loss is not the lever, reconfirms low-energy HG gap is capacity-bound, no reweighting can fix at < 100 parameters.

Distillation Exp 2: Loss Shaping (negative result)



Relative-loss HG band (much wider)
next to the from-scratch band

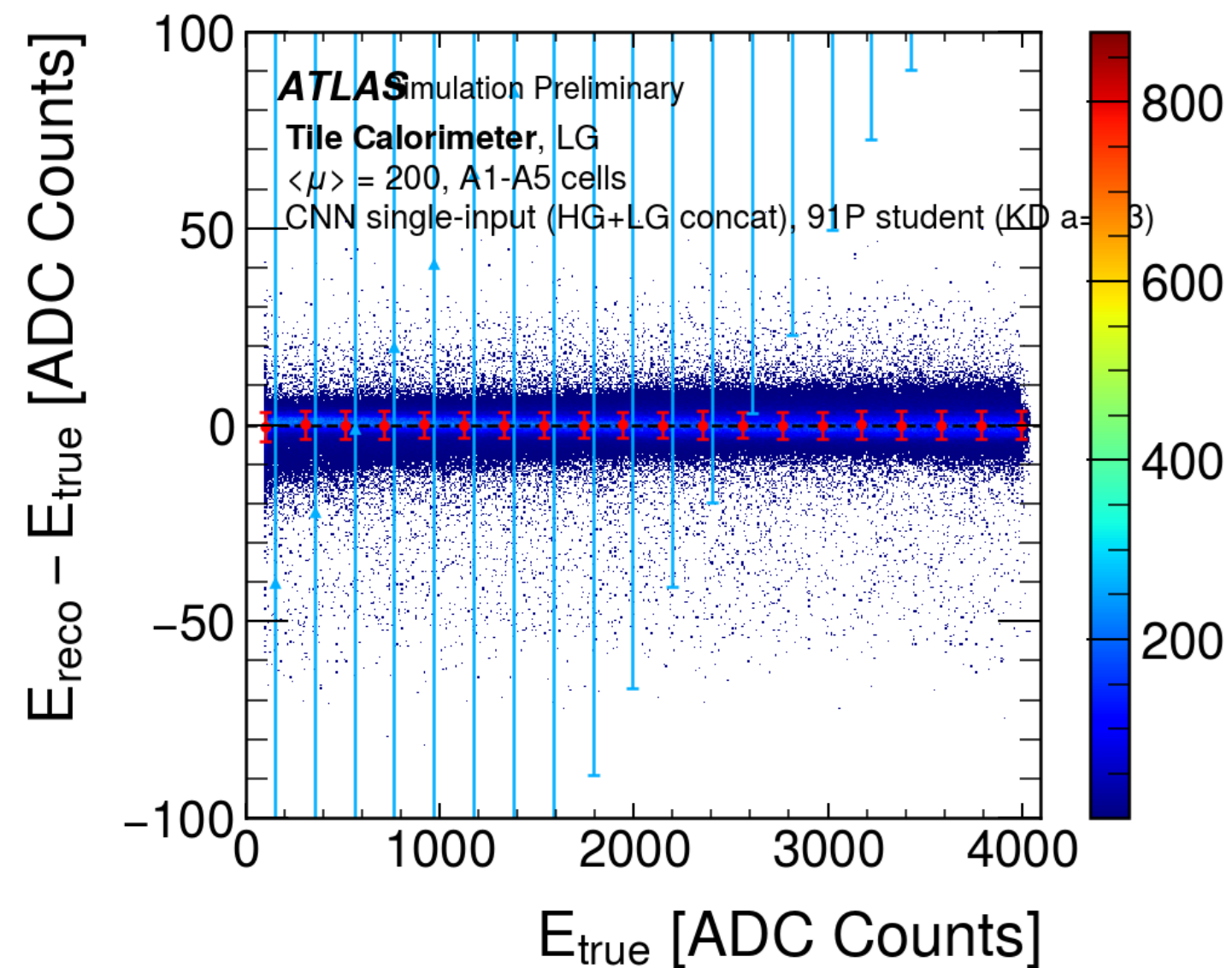
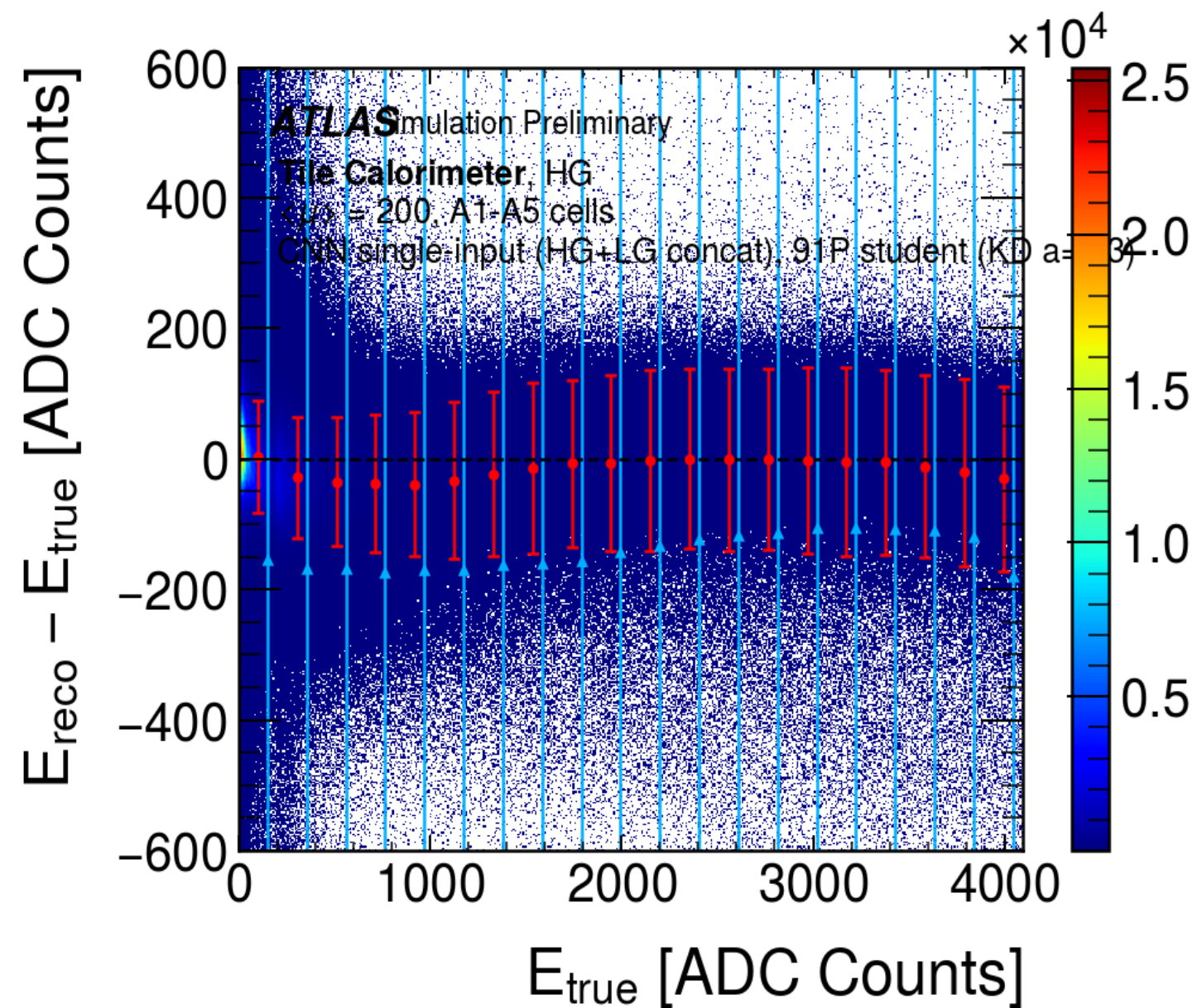
All Sub-100P Student Experiments (full matrix)

Best student:- KD alpha=0.3 (91P) | Conv1d(1->4,k5) -> Conv1d(4->3,k3) -> Linear(27->1), LeakyReLU(0.25)

Run	Params	Distillation	HG sig (global)	HG sig (bin-mean)	LG sig
From-scratch (no teacher)	91	none	105.9	164.8	4.72
KD alpha=0.7 (truth-heavy)	91	light, MSE	103.7	148.3	4.14
KD alpha=0.5	91	light, MSE	96.4	128.5	3.89
KD alpha=0.3 (best)	91	light, MSE	95.4	127.0	3.43
KD alpha=0.1, teacher 533P	99	MSE	104.8	138.4	4.17
KD alpha=0.1, teacher 1005P	99	MSE	96.6	148.2	4.50
KD alpha=0.3, scale-fixed	99	heavy, RMSE+MAE	115.9	194.4	5.90
KD alpha=0.5, scale-fixed	99	heavy, RMSE+MAE	116.4	184.4	5.73
Relative-resolution loss	91	none, rel loss	109.8	212.6	7.50
-- anchors --					
Teacher (ours)	1005	--	63.0	85.8	2.70
IFIC reference CNN (HG-only)	147	--	72.12	flat-ish	8.36

- Light KD (alpha=0.3) is the best at -95 HG, heavier teacher-matching (scale-fixed) AND relative-resolution loss are both worse — the two dead ends.
- The 91P student does BOTH gains and beats the reference on LG (-2.5x), HG trails at the capacity floor. The reference 72 is a 147P HG-only model.

All Sub-100P Student Experiments



Conclusions

- Teacher:- a calibrated 1005-parameter CNN, merged (ificuw/tilecal!16), HG resolution comparable to the reference, LG clearly better (~2.5x), near the pile-up noise floor.
- **Student**:- a working sub-100-parameter distilled model (91P, same input/output and calibration as the teacher). Beats the reference on LG, on HG it reaches ~95 ADC with the approaches explored so far.
- **Distillation behaviour mapped**:- a light teacher weight helps (~10% over from-scratch), leaning too hard on the teacher hurts, useful to carry forward.
- The remaining HG gap to the teacher looks tied to the <100-parameter budget – open to input on where to push next.